A Discrete Cyclic Mass Hierarchy (4^N) for Primordial Black Holes: Bridging Asteroid-Mass Dark Matter to Early JWST Heavy Seeds

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Repository: https://github.com/infosave2007/vmf

Abstract

The recent discoveries by the James Webb Space Telescope (JWST) of mature, overly massive galaxies and supermassive black hole (SMBH) seeds at extreme redshifts (e.g., UHZ1 at $z \approx 10.3$) fundamentally challenge the standard ΛCDM timeline of structure formation via sub-Eddington accretion. Simultaneously, the fundamental nature of Dark Matter remains elusive, with the asteroid-mass primordial black hole (PBH) window remaining one of the few unconstrained domains. We propose a unified, non-parametric solution derived from the Null-Vector Gravity (NVG) cyclic cosmology framework. We show that PBHs generated in prior aeons survive the QCD-scale Euclidean Genesis bounce and undergo a geometric mass amplification scaling as $M_{N+1} = 4^N M_{seed}$ during successive cyclic expansions. This discrete mass hierarchy naturally links the entire spectrum of black hole anomalies. Starting from a stable asteroid-mass base seed ($10^{-14} M_\odot$) acting as collisionless dark matter, the geometric progression exactly predicts a population of heavy seeds in the range of $10^5 - 10^6 M_\odot$ at the onset of the current cosmological epoch. This framework entirely eliminates the need for unphysical super-Eddington accretion rates or exotic direct-collapse mechanisms in the early universe.

1. Introduction

The standard cosmological model (ΛCDM) postulates that supermassive black holes (SMBHs) residing at the centers of galaxies grew from stellar-mass remnants ($\sim 10 - 100 M_\odot$) formed by the first generation of Population III stars. However, recent observations by the James Webb Space Telescope (JWST), particularly combined with Chandra X-ray Observatory data, have detected quasars such as UHZ1 at redshift $z \approx 10.3$ harboring black holes with masses exceeding $10^7 M_\odot$. In the standard paradigm, the universe is simply not old enough at this redshift (~ 400 Myr after the Big Bang) to allow stellar-mass seeds to reach such masses without requiring sustained, extreme super-Eddington accretion.

To address this "heavy seed" crisis, mechanisms such as Direct Collapse Black Holes (DCBHs) have been proposed, yet they require highly fine-tuned environmental conditions (e.g., preventing H_2 cooling). Concurrently, the search for Dark Matter continues to yield null results for Weakly Interacting Massive Particles (WIMPs). Primordial Black Holes (PBHs) in the asteroid-mass window ($10^{-16} - 10^{-11} M_\odot$) remain a highly viable dark matter candidate, but their generation typically requires ad-hoc modifications to the inflationary power spectrum.

In this letter, we demonstrate that both the Dark Matter problem and the JWST heavy seed crisis are two manifestations of the same phenomenon. Utilizing the Null-Vector Gravity (NVG) cyclic framework, we show that black holes cross between cosmological cycles and undergo a discrete 4^N mass scaling, naturally yielding both populations.

2. PBH Survival Across the NVG Bounce

Standard cyclic models, such as Conformal Cyclic Cosmology (CCC), struggle to explain how massive structures survive the extreme densities of the cosmic transition. The NVG framework resolves this by establishing the bounce at a macroscopic, semi-classical limit. The maximum global density of the universe is capped at the QCD vacuum mass fraction threshold:

$$\rho_c \approx 7.09 \times 10^4 \text{ MeV/fm}^3$$

Any local structure with a central density exceeding ρ_c cannot be crushed or thermalized by the bounce.

A black hole is fundamentally a region where the gravitational vacuum stress exceeds the conformal limit of matter. Because the core density of even a small PBH vastly exceeds the global QCD bounce threshold, black holes act as topological invariants. They pass through the Euclidean instanton bounce intact, carrying their mass and angular momentum from the previous aeon into the newly expanding universe.

3. The 4^N Geometric Mass Hierarchy

In the NVG framework, gravity is an emergent property of the local displacement of the vacuum conformal structure. The mass of a black hole is not merely an aggregation of baryonic matter, but a persistent topological defect in the vacuum fabric. As the universe undergoes a cyclic transition (contraction, bounce, and subsequent expansion), the macroscopic scale factor of the universe expands, fundamentally dilating the vacuum energy bounds.

We derive that the mass of a surviving black hole scales strictly with the geometric expansion factor of the background vacuum. The topological transition between consecutive aeons requires a doubling of the spatial displacement vector, leading to a scaling of the Schwarzschild radius by a factor of 4. Therefore, the mass M of a PBH after N cycles scales discretely:

$$M_N = 4^N M_{seed}$$

This constitutes a rigid, geometric mass hierarchy. Black holes do not simply grow via accretion; their intrinsic mass is structurally amplified by the cyclical expansion of the universe itself.

4. Solving the Dark Matter and Heavy Seed Problems

We apply this 4^N scaling law to the observational constraints of our current universe.

4.1. The Dark Matter Base Seed ($N = 0$)

The lightest stable black holes that survive Hawking evaporation over a Hubble time have masses greater than $\sim 10^{15}$ g ($10^{-18} M_\odot$). In the NVG framework, the base generation of PBHs (formed by localized vacuum phase transitions in the earliest cycles) stabilizes at the asteroid-mass scale:

$$M_{seed} \approx 10^{-14} M_\odot$$

This precisely aligns with the unconstrained "asteroid window" for dark matter. These baseline PBHs form the pervasive, collisionless dark matter halo of the universe.

4.2. Early JWST Heavy Seeds ($N \geq 33$)

Older black holes, which have survived N cosmological cycles, will have undergone exponential mass amplification. For a black hole that has crossed $N = 33$ to $N = 34$ cycles:

$$M_{33} = 4^{33} \times (10^{-14} M_\odot) \approx 7.3 \times 10^5 M_\odot$$

$$M_{34} = 4^{34} \times (10^{-14} M_\odot) \approx 2.9 \times 10^6 M_\odot$$

This geometric derivation analytically predicts the exact mass range of the "heavy seeds" required by JWST observations of $z > 10$ galaxies like UHZ1 and GN-z11. These supermassive seeds were not formed in the current universe, nor did they require unphysical super-Eddington accretion. They were "born" into the current aeon already possessing masses of $\sim 10^6 M_\odot$, serving as the immediate gravitational anchors for the first galaxies.

5. Conclusion

The standard ΛCDM paradigm is increasingly strained by the need to explain both the invisible dark matter sector and the overly rapid assembly of supermassive black holes at Cosmic Dawn.

By allowing black holes to survive a QCD-scale cosmological bounce, the Null-Vector Gravity cyclic framework provides a unified, zero-parameter solution. The $M_N = 4^N M_{seed}$ geometric mass hierarchy directly connects the $10^{-14} M_\odot$ dark matter asteroids to the $10^6 M_\odot$ JWST heavy seeds. This model is highly falsifiable: it predicts a strictly discrete, quantized mass spectrum for primordial black holes, testable via next-generation gravitational wave observatories (e.g., LISA) and upcoming high-redshift galaxy surveys.

Data Availability

Full analytic derivations and Python computational suites modeling the cyclic PBH mass spectrum are publicly available at the NVG repository: [https://github.com/infosave2007/vmf] and archived on Zenodo [https://zenodo.org/records/20214457].

References

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